MEASURING BENEFITS OF ENVIRONMENTAL IMPROVEMENTS

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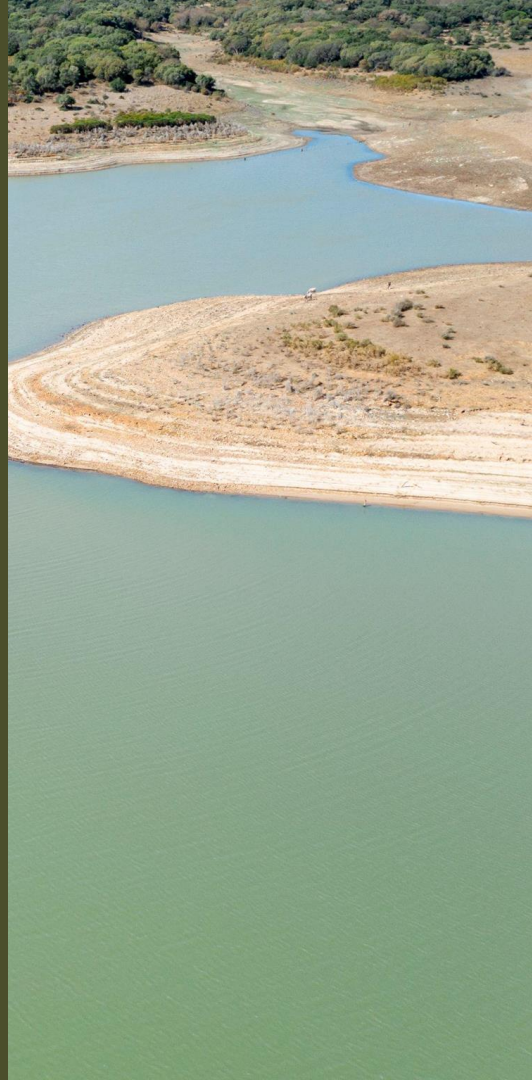


01 Economic Value of Environment

Economic Value of Environment

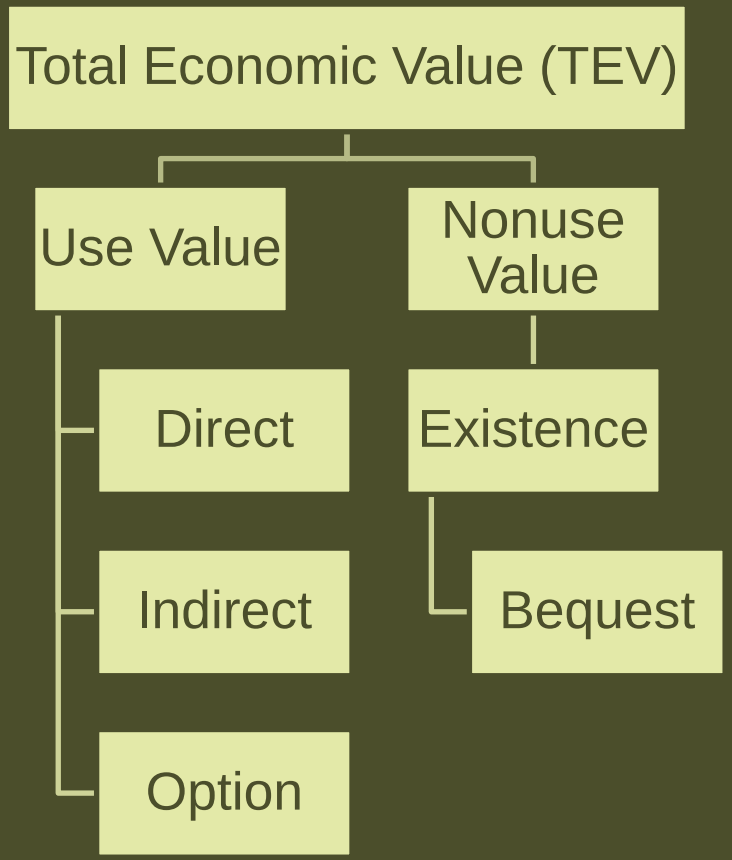
Objective: To obtain a measurement of the benefit to the welfare of individuals and the society or cost to incur due to environmental degradation by the individuals and the society in monetary terms.

Goal: is not just to measure the benefit or cost in money terms but help the policy makers in their policy initiatives regarding allocation of resources, imposition of taxes, various compensation schemes such as subsidies.

Economic Value of Environment

To measure the benefits and costs in monetary terms environmental valuation therefore considers the use of environmental goods, improving the conditions of environmental goods and damage, if any, and remedial measures to correct environmental damage.

Economists have developed tools to measure environmental values. For this it estimates individual's willingness to pay to benefit from environmental goods. The costs due to environmental damage are measured by the loss suffered by the individuals who benefited from the damaged good and deciding the appropriate compensation for losing the benefit (willingness to accept).


Total Economic Value (TEV)
Use Value
Nonuse Value
Direct
Indirect
Option
Existence
Bequest

https://www.youtube.com/feed/downloads

02

Measurement Methods of Environmental Value

Measurement Methods of Environmental Value

Objective Standard Based Valuation

Direct market valuation approach which uses data on actual markets

Subjective Preference Based Valuation

Revealed preference approach where people indirectly reveal their preferences for the environmental good through their choice of the marketed good.

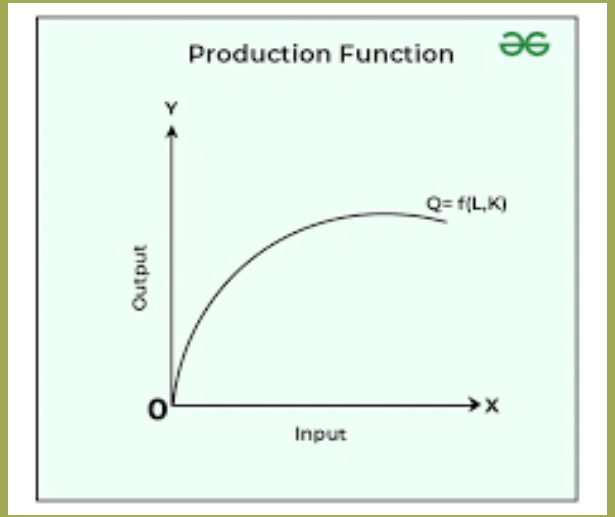
Stated Preference Method

Hypothetical markets are constructed and people are asked to state their willingness to pay for the environmental good.

Objective Standard Based Valuation

- Benefit
  - Dose-Response
  - Foregone Earning Approach
- Cost
  - Opportunity Cost
    - Relocation Cost
      - Replacement Cost

Dose-Response/Production function approaches



- An increase in output due to the change is a measure of an increase in benefit
- A decrease in output is a measure of an increase in cost.
- Production function based approach analyses the effect of pollution externalities on production function i.e. here the environmental impact on inputs and ultimately on output is traced

Foregone Earning /Human Capital Approach

- The human capital approach values environmental attributes through their effects on the quantity and quality of labour.
- The principle involved in this approach is that of valuing life in terms of the value of labour.
- Given adequate data regarding lifetime earnings, participation rates in the labour force mortality rates, etc., it is possible to estimate the value of the expected future earnings of individuals in any age-group.

Opportunity Cost Method

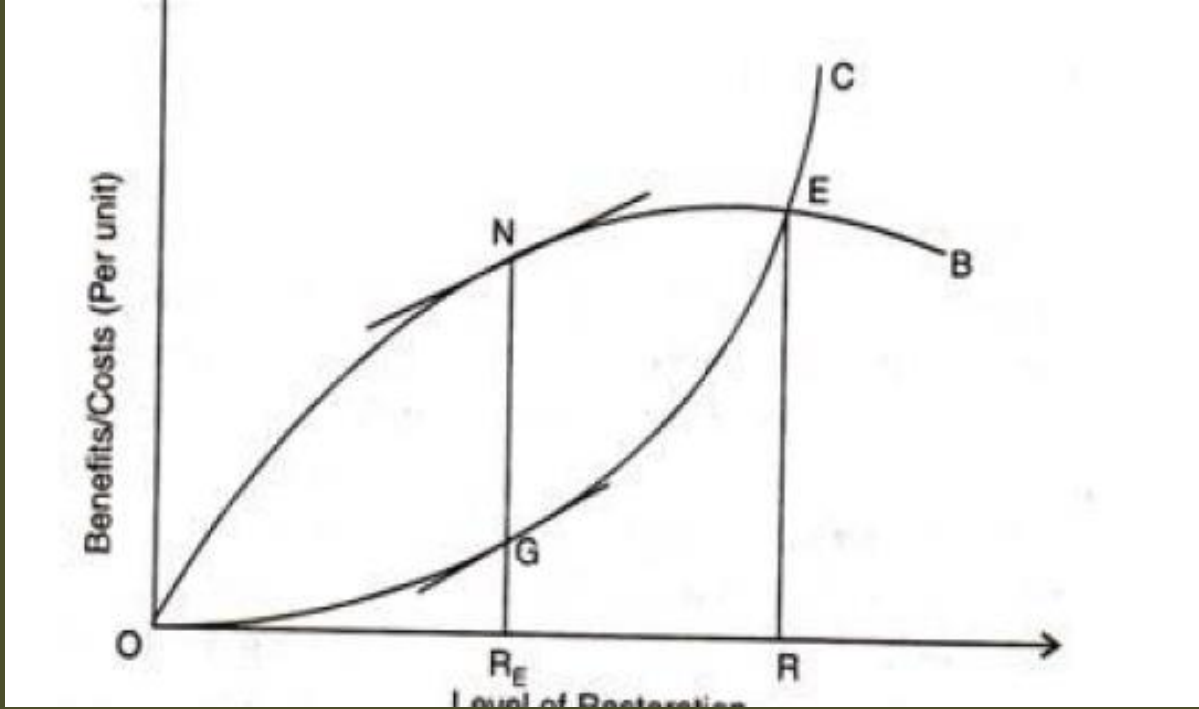
- This method values the benefits of environmental protection in terms of what is being foregone to achieve it.
- The opportunity cost method is useful in cases where it is difficult to enumerate the benefits of an environmental change

Relocation Cost Method

This is a cost-based technique used to estimate the monetary value of environmental damages based on the potential costs of relocating a physical facility that would be damaged by a change in environmental quality. This method relies on data on potential expenditures

Replacement Cost Method

- This is a Cost-based technique that measures the potential expenditures that would be required to replace or restore a productive asset that would be damaged by some project or development.
- These costs are then compared to the costs of preventing the damage from occurring to determine which is more efficient.



The slope of curve B indicates that with the increase in restoration level, benefits increase at a decreasing rate. The slope of curve C indicates that the restoration costs are an increasing function of the level of restoration.

The economic efficiency is achieved at the restoration level ORE where the difference between curve B and curve C is the maximum. The net gain is NG at this level of restoration.

Subjective Preference Based Valuation

- Hedonic Price Method
- Travel –Cost Method

Hedonic Price Method

Assumption: Price of a property is related to the stream of benefits to be derived from it.

Hypothesis: The prices which individuals pay for commodities reflect both environmental and non-environmental characteristics.

Environmental	**Non-environmental**
Cleanliness	Age of the house
Pollution Level	Size
Ambience	Material Used
Commute Option	Maintenance

10000 Sq Ft

Outskirts of the city

Price: 70 Lakhs

Sea Facing

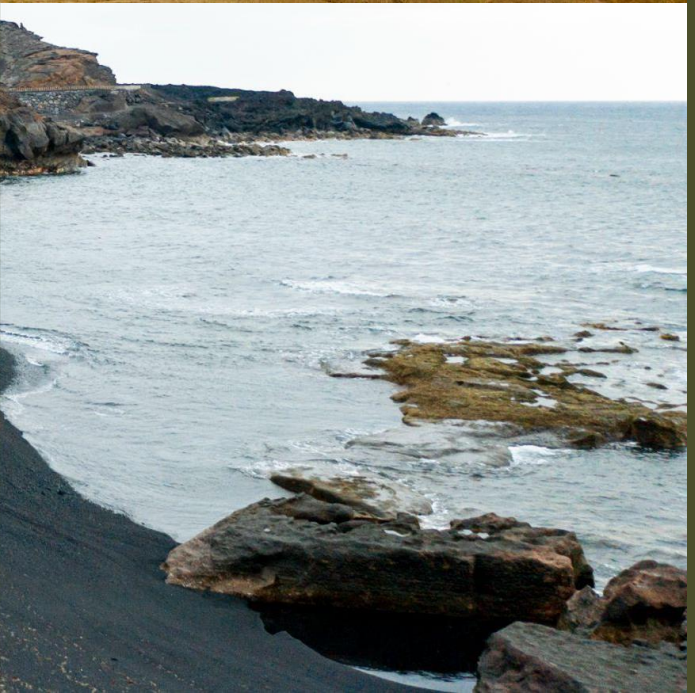
Price: 90 Lakhs

Sea Facing, Hospitals, School, IT Park

Price: 1 cr 10 Lakhs

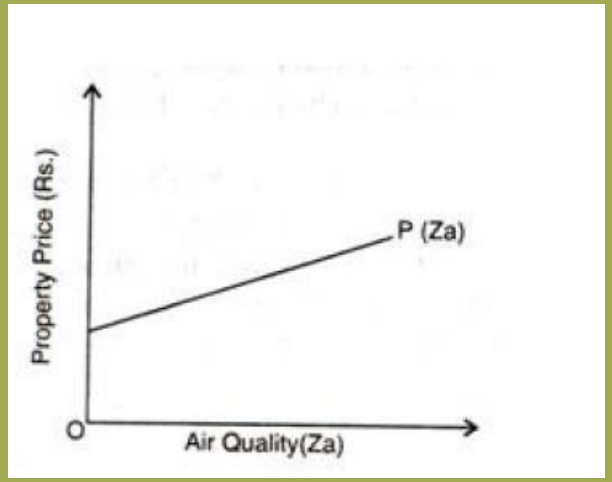
Hazardous Chemical

Price: 50 Lakhs



EMERGENCY
SCHOOL

Hedonic Price



P = f(S, N, Z)

Where S is structural characteristics: size, rooms, age.
N is neighborhood characteristics: Distance from school, hospital, crime rate.
Z is environmental characteristics: air quality.

Hedonic Price Method

The hedonic price approach attempts to identify how much of a property differential is due to a particular environmental difference between properties, and how much people are willing to pay for an improvement in the environmental quality that they face and what the social value of improvement is.

Travel Cost Method

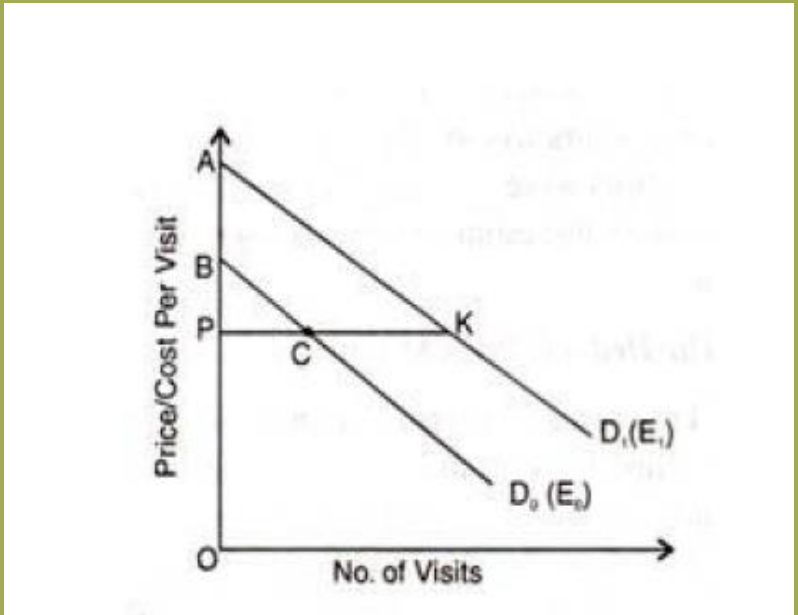
The travel-cost method is a widely used surrogate market approach that relies on information on time and travel costs to derive a demand curve for a recreational site.

This curve is in turn used to estimate the consumers' surplus or value of the site to all users.

This approach is widely used to value the recreational benefits of public parks and other natural areas.

Dizzee World

Travel Cost



D=f(price, visitors' income, and socio-economic Characteristics).

D is Demand for a recreational site (i.e. number of visits per year to a park)

The price is usually, cost of travel, and opportunity cost of time spent.

The consumers' surplus associated with the demand curve provides an estimate of the value of the recreational site in question.

Stated Preference Based Valuation

Contingent Valuation Method (CVM)

Contingent Valuation Method (CVM)

This method puts direct questions to individuals to determine how much they might be willing to pay for environmental resources or how much compensation they would be willing to accept if they were deprived of the same resources.

I would pay
Rs_____

A

B

I would pay RS_____ to turn B to A

Would you pay Rs 1000 per year to keep this around?

YES NO

I would pay

_____

Rs 100
Rs 300
RS 500
Rs 800

I would accept

_____

to compensate for this

Trade-off Game Method

Which one do you choose?

Costless-Choice Method

Delphi Method

The Delphi Method is also known as Expert Opinion method. In this survey method, opinion of the experts have been taken into consideration.

By using their expertise in respective field, experts place values on a nonmarketed good or service through a repetitive process where feedback among the group is considered between each step of repetition.

Summary

Topic Discussed

Economic value of environment; use and non use value; measurement method of environment value; objective valuation; subjective preference valuation; stated preference valuation; production function method; human capital method; opportunity cost method; relocation cost; replacement cost; Hedonic price method; travel-cost method; contingent valuation method.

Assignment

NA

Activities

NA

Thank You!

Do you have any questions?

“End of Education is Character”